Man Singh

B.Tech, Computer Science & Engineering
Indian Institute of Information Technology, Agartala

maansimaan02@gmail.com
linkedin.com/in/maan-s-978136329
github.com/man-singh-dev

Education

Degree	Institute	Score	Year
B.Tech (CSE)	Indian Institute of Information Technology, Agartala	8.82 / 10	2024 – 2028
Class XII	Modern English School, Nawada, Bihar	94.20%	2023
Class X	Modern English School, Nawada, Bihar	98.60%	2021

Experience

Indian Institute of Technology Patna Aug 2026 – Present
AI/ML Research Intern, under Dr. Sriparna Saha *Vision-Language Models*

- Led a student research team studying whether Vision-Language Models genuinely reason about football tactics (*why* an action succeeds or fails) or only describe what is visible on screen.
- Established baselines for open and proprietary VLMs on a ∼1,000-question football benchmark covering temporal order, causality, space creation and passing lanes.
- Extended the benchmark with harder temporal, causal and counterfactual questions designed to separate tactical reasoning from visual description.
- Verified model explanations against player-tracking data; compared performance across tracking data, tactical diagrams and broadcast video.
- Reviewed literature on video QA and sports benchmarks to identify the research gap; proposed RL with rewards from tracking-verified tactical claims.

Crukx Jun 2026 – Jul 2026
Frontend Developer

- Redesigned the company website's UI (layouts, hierarchy, styling) and built it in TypeScript, JavaScript, HTML/CSS.
- Reworked page structure and UI components so the whole site follows one consistent design language.
- Fixed responsive layouts so pages work properly across screen sizes and devices.

Projects

SentinelOps | Event Ingestion Platform *Go, PostgreSQL, Docker, REST*

- Built a backend in Go where services register and send events via REST; runs in Docker, tested end-to-end over HTTP.
- API keys are hashed with SHA-256 before storage and verified against PostgreSQL on each request; request bodies are validated before ingestion.
- Made ingestion idempotent with a unique constraint and conflict-safe inserts. Firing 25 identical requests at once stores exactly one row.
- Wrote Go tests for auth, validation, duplicate events and concurrent requests.

DiffCraft | Collaborative Code Review Platform *Next.js, Fastify, PostgreSQL, Redis, BullMQ, Socket.IO*

- Built a code review tool with line-level comments, threaded discussions and review approvals: Next.js frontend, Fastify + Drizzle/PostgreSQL backend, with auth and access control.
- Comments, review status and notifications sync live between reviewers over Socket.IO.
- Added AI review: code is queued through BullMQ/Redis and sent to the OpenAI API, which returns structured findings (likely bugs, code smells, security issues) with suggested fixes.

Technical Skills

Languages	C++ (DSA), Java, Go, Python, TypeScript, JavaScript, SQL, HTML/CSS
Web & Backend	Node.js, Fastify, Next.js, React, REST APIs, Socket.IO, BullMQ
Databases	PostgreSQL, Redis, Drizzle ORM
AI / ML	Vision-Language Models, Video Understanding, Benchmark Design, Model Evaluation, OpenAI API
Tools	Git, GitHub, Docker, Vitest, Zod
Coursework	DSA, OOP, DBMS, Operating Systems, Computer Networks

Achievements

- LeetCode: 1800+ rating, 600+ problems solved | Codeforces: 1287 rating, 300+ problems solved.
- Semi-Finalist, ET Gen AI Hackathon; selected for Smart India Hackathon (SIH) at the institute level.

Position of Responsibility

E-Cell, NIT Agartala | *Lead, Technical & PR Team*

- Handled tech and PR for E-Cell events; worked with sponsors, participants and organizing teams on planning and outreach.